Harsha Vardhan Rapeti

+91 7842751810 rharshavardhan2004@gmail.com LinkedIn Github

Technical Skills

Languages C++, Python, Rust, Bash, TypeScript, MySQL, Ruby, Faust
Frameworks FastAPI, Ruby on Rails, JUCE, LangChain, React, TensorFlow
Developer Tools Git, CMake, Make, Visual Studio, GDB, Docker, Valgrind, Perf, Postman

Professional Experience

Software Developer Intern **Aug 2024 – May 2025**
Pitch Innovations

- Worked on **Fine-Tuning Llama based MIDI generation models** for different generes of music.
- Containerized and shipped PyTorch models using **ONNX** for inference on client machine.
- Developed **VST and standalone audio plugins** for Windows and macOS using **C++ and JUCE framework**, deployed to **1,000+ active users**.
- Architected **cross-platform build systems** using CMake(for Windows/Mac/Linux) and Integrated **Faust DSP language** into build pipeline, accelerating audio effect **development cycle by 60%** through automated code generation.
- Implemented **8+ professional-grade audio effects** including delays, compressors, distortions, Schroeder reverb, granular synthesis, and convolution reverb with **optimized real-time performance**.
- Technologies: PyTorch, Tensorflow, VS2022, CMake, GDB, perf, JUCE, docker, Git, Python, C++

Projects

Warmer - Node-Based Audio Synthesizer repo
Open-source audio synthesis platform with 25+ unique nodes and real-time graph processing

- Engineered **lock-free multi-threaded audio processing engine** using topological sorting algorithms, supporting **graphs with 100+ nodes at 48kHz sample rate**
- Implemented **15+ DSP algorithms** including FFT-based spectral processing, IIR/FIR digital filters, and non-linear waveshapers.
- Designed efficient **XML-based serialization system** for graph state persistence.
- Technologies: C++17, CMake, Make, Mingw-w64, JUCE, GDB, Valgrind, Git

Analytiks - Audio Visualization Software repo **May 2025 – Jun 2025**
Professional audio analysis plugin with OpenGL-accelerated visualizers

- Developed **4** real-time audio visualizers (Correlation Meter, Spectrogram, Oscilloscope, Spectrum Analyzer) using OpenGL shaders, achieving **60+ FPS at 1080p resolution**
- Optimized inter-thread communication using **non-blocking lock-free queues**.
- Automated build and distribution pipeline using GitHub Actions CI/CD for Windows, OSX and Linux.
- Technologies: C++, CMake, Visual Studio 2022, JUCE, OpenGL, Git, GitHub Actions

SynthBite16 - 16-bit Computer Architecture repo **Nov 2023 – Jan 2024**
Fully functional 16-bit computer design with custom instruction set architecture

- Designed **complete 16-bit computer architecture** with 38 instructions, 2 addressing modes, and hardware register stack in Logisim
- Optimized instruction format using 4-nibble encoding scheme, achieving **95% opcode utilization efficiency**
- Implemented **Harvard architecture** with separate 132KB instruction and data memory spaces
- Created **custom assembly language and assembler in C++**, supporting symbolic addressing
- Technologies: Logisim, C++, Assembly Language Design, Computer Architecture

Education

BIT Mesra (Birla Institute of Technology) **Oct 2022 – Present**
Bachelor of Technology in Artificial Intelligence and Machine Learning
CGPA: 7.61/10.0

Presentations and Talks

ADC India 2025: Spatial Audio Localization Techniques **Conference Presentation**
Presented at Audio Developer Conference (ADCx) India Watch

- Delivered technical presentation on implementing real-time binaural audio synthesis using Head-Related Transfer Functions (HRTFs)
- Presented to audience of **100+ audio developers, engineers, and industry professionals**